Volume I · Chapter 0 — Python & GitHub Fundamentals · Homework

> Onboarding chapter (added after V1), maintained independently. Hour model **1 : 2 : 4** (theory : labs : homework) — this is the **4×** independent-practice component. All homework is **submitted through the student's own GitHub repository** (commits + at least one merged Pull Request), so Git/GitHub practice is built into the deliverables. See `Volume-I-Chapter-0-context.md`.

Prerequisites: Chapter 0 Theory + Labs.

Learning outcomes (homework): the student can learn independently from curated resources, solve a broad set of exercises, build and publish a small project on GitHub with a professional workflow, and read/apply Python best practices.

Homework — 40 h:

Python (28 h)

#	Assignment	h	Primary resources
HW 0.1	Work a structured beginner track (e.g., first ~10 days of **30-Days-Of-Python**, or the W3Schools Python track + selected Python.org tutorial chapters); keep daily notes	8	W3Schools Python; Python.org tutorial; 30-Days-Of-Python
HW 0.2	Solve **20–30 exercises** (beginner → intermediate) from a practice set	8	*Python Programming Exercises*; W3Schools exercises
HW 0.3	Build **one small project from scratch** (CLI app / simple game / data parser)	8	**Project-Based Learning**; *Amazing Python Scripts*
HW 0.4	Read & apply best practices (RealPython	4	The Hitchhiker's Guide to Python; Awesome Python

#	Assignment	h	Primary resources
	"Hitchhiker's Guide"); refactor your project; pick one library from **Awesome Python** and write a short demo		

Git & GitHub (12 h)

#	Assignment	h	Primary resources
HW 0.5	Complete the **W3Schools Git** tutorial + GitHub's **Git & GitHub learning resources**; write a one-page cheat-sheet	4	W3Schools Git; GitHub learning resources
HW 0.6	Publish your HW 0.3 project to GitHub: proper **README**, `.gitignore`, meaningful commit history, and a feature branch merged via **Pull Request**	4	GitHub *Hello World*; GitHub docs
HW 0.7	Practice the collaboration flow: fork a reference repo (or a partner's), open an **issue**, submit a small **PR**, and document the round-trip	4	GitHub docs; Class Central Git/GitHub courses

Submission & assessment: each assignment is submitted as part of the student's GitHub repo — graded on completeness, correctness, **commit hygiene**, README quality, and at least **one merged Pull Request**. Suggested weighting: Python HW **60%**, Git/GitHub HW **40%**.

Key decisions: choosing which curated track to follow; scoping a first project realistically; structuring a README; branch-per-feature vs. commit-to-main for solo work.

References & learning resources:

- W3Schools Python — https://www.w3schools.com/python/
- Python.org tutorial — https://docs.python.org/3/tutorial/index.html
- W3Schools Git — https://www.w3schools.com/git/
- GitHub "Hello World" — https://docs.github.com/en/get-started/using-github/hello-world
- GitHub — Git & GitHub learning resources — https://docs.github.com/en/get-started/start-your-journey/git-and-github-learning-resources
- Class Central — best Git/GitHub courses — https://www.classcentral.com/report/best-git-github-courses/
- GitHub repos (verify current URLs): `github.com/Asabeneh/30-Days-Of-Python`, `github.com/practical-tutorials/project-based-learning`, `github.com/TheAlgorithms/Python`, `github.com/vinta/awesome-python`, `github.com/geekcomputers/Python`, `github.com/realpython/python-guide`; plus *Learn-Python*, *Python Programming Exercises*, *Amazing Python Scripts*, *Learn Python 3* (locate on GitHub).
- Course tracks: *Python for Data Science, AI & Development* (IBM/Coursera), *Google IT Automation with Python*, *IBM AI Developer*.

Note: original source links were provided as `lnkd.in` short-redirects; canonical paths above should be re-verified before publishing.

Hours: Homework **40** (paired with Theory 10 and Labs 20; ratio 1 : 2 : 4).